

Experiments

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

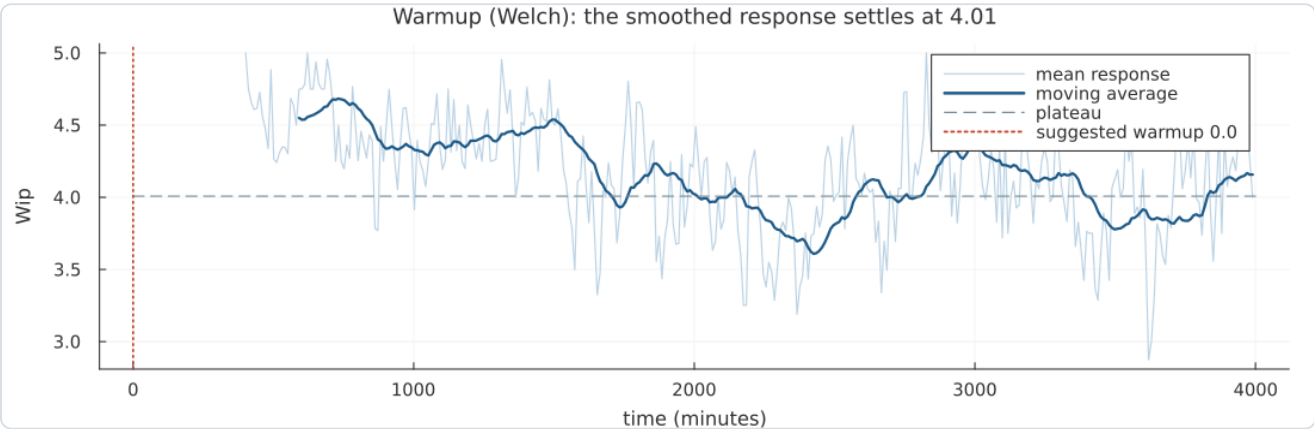
Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 | Generated: | Engine: DiscreteSim | Sections: 8

Experiments

Replications, a detected warmup, batch means for a single run, paired comparisons and a factorial design -- all with common random numbers.

How long the transient lasts

Welch's procedure on the wip series over 4 replications suggests discarding the first 0 time units (plateau 4.008 ± 0.096).



Warmup

One long run, honestly

Batch means

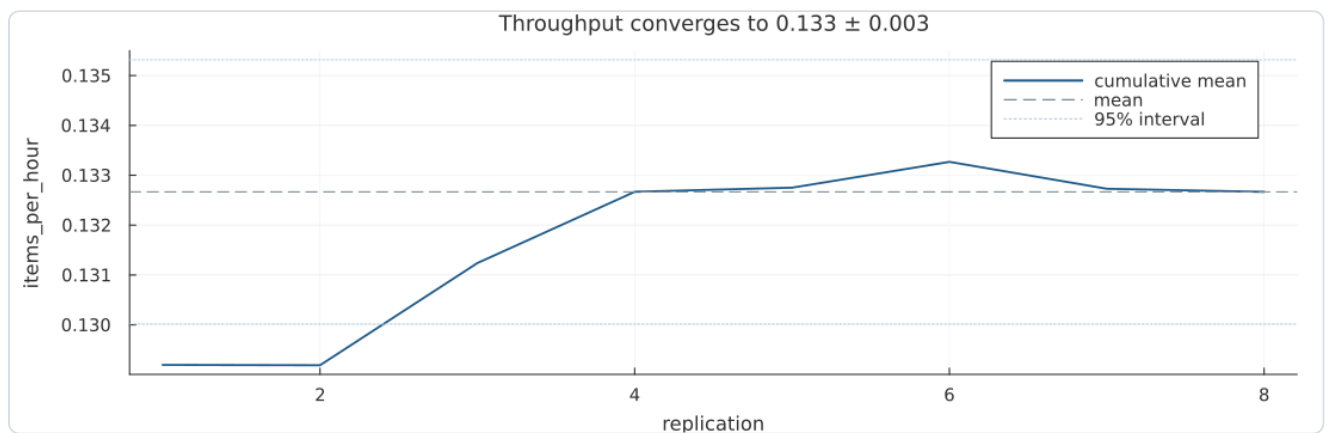
N	Mean	Half width	Lo	Hi	Batches	Lag1	Adequate
8	552.5	8.555	544	561.1	4	-0.543	no

The replications

Machine shop over 8 replications

Metric	Mean	Half width	Unit	Rel hw pct
wait_mean	552.5	12.7	minutes	2.294
wait_p95	1034	103.3	minutes	9.987
cycle_time_mean	552.5	12.7	minutes	2.294
service_mean	—	—	minutes	—
completed	477.5	9.543	count	1.998
throughput	0.133	0.00265	items_per_hour	1.998
wip_mean	4.155	0.061	count	1.476

Metric	Mean	Half width	Unit	Rel hw pct
tardiness_mean	449	11.5	minutes	2.564
scrapped	22.9	4.057	count	17.7
reworked	71.2	5.031	count	7.061
setup	2	0	minutes	0
utilisation	0.863	0.011	ratio	1.285
availability	1	0	ratio	0
max_queue_length	44.1	6.49	count	14.7
queue_length_mean	75.5	0.054	count	0.071
span	3599	0.388	count	0.011
events	7762	105.8	count	1.363



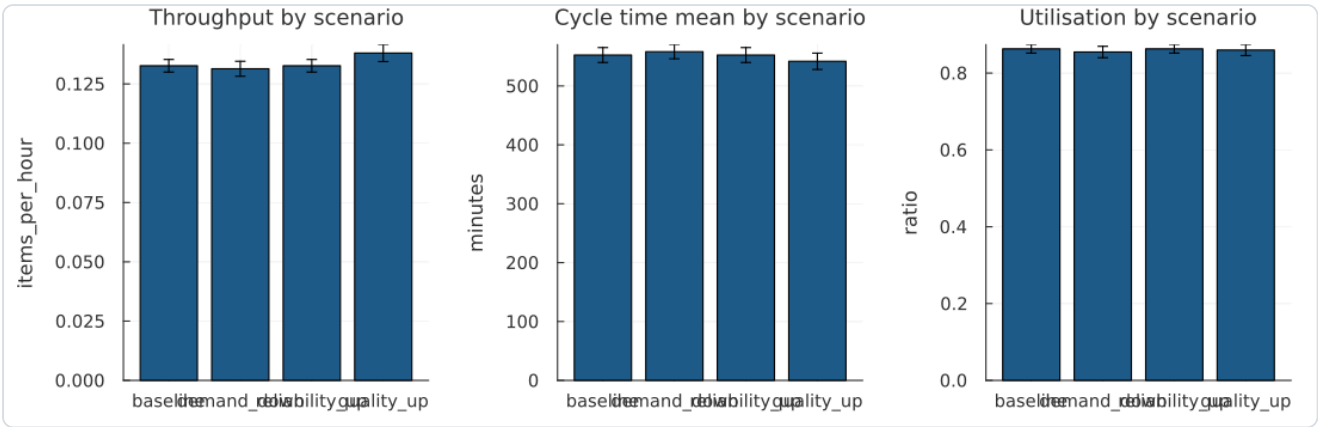
Convergence

Scenarios against the baseline

Scenarios

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
demand_down	throughput	0.133	0.131	-0.00126	0.00163	-0.953	indistinguishable
demand_down	cycle_time_mean	552.5	558.2	5.679	3.709	1.028	worse
demand_down	wait_mean	552.5	558.2	5.679	3.709	1.028	worse
demand_down	utilisation	0.863	0.855	-0.00824	0.00893	-0.955	indistinguishable
demand_down	completed	477.5	473	-4.5	5.844	-0.942	indistinguishable
reliability_up	throughput	0.133	0.133	0	0	0	indistinguishable
reliability_up	cycle_time_mean	552.5	552.5	0	0	0	indistinguishable
reliability_up	wait_mean	552.5	552.5	0	0	0	indistinguishable
reliability_up	utilisation	0.863	0.863	0	0	0	indistinguishable
reliability_up	completed	477.5	477.5	0	0	0	indistinguishable
quality_up	throughput	0.133	0.138	0.00535	0.00154	4.032	better
quality_up	cycle_time_mean	552.5	541.9	-10.6	5.537	-1.926	better
quality_up	wait_mean	552.5	541.9	-10.6	5.537	-1.926	better

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
quality_up	utilisation	0.863	0.86	-0.00318	0.00578	-0.369	indistinguishable
quality_up	completed	477.5	496.8	19.2	5.523	4.031	worse



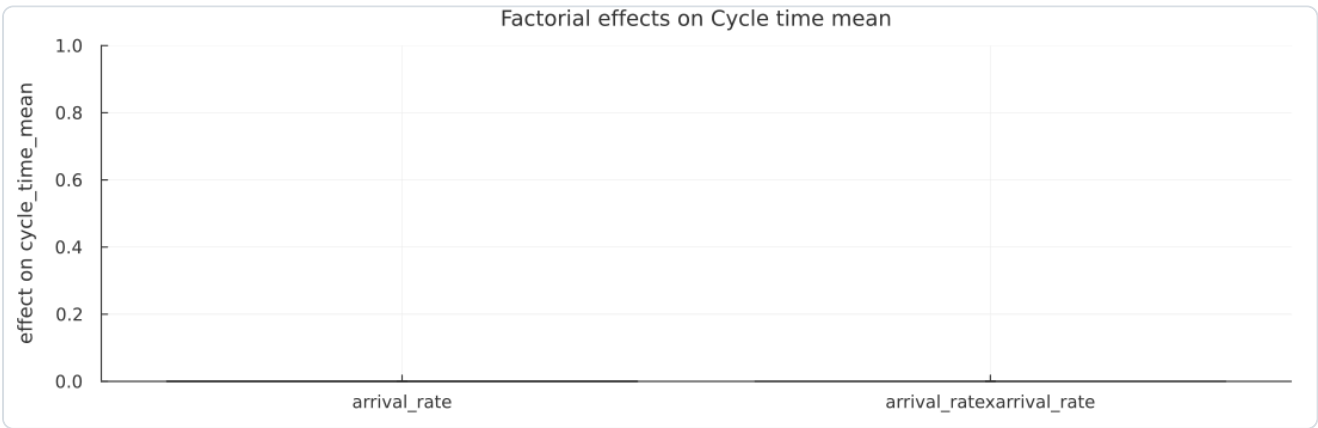
Scenarios

Best by cycle_time_mean: **quality_up** (compared with baseline, common random numbers on every replication).

Which knob matters?

Factorial

Term	Metric	Effect	Half width	Lo	Hi	Verdict
arrival_rate	cycle_time_mean	0	0	0	0	negligible
arrival_rate	cycle_time_mean	0	0	0	0	negligible
arrival_ratearrival_rate	cycle_time_mean	0	0	0	0	negligible



Factorial